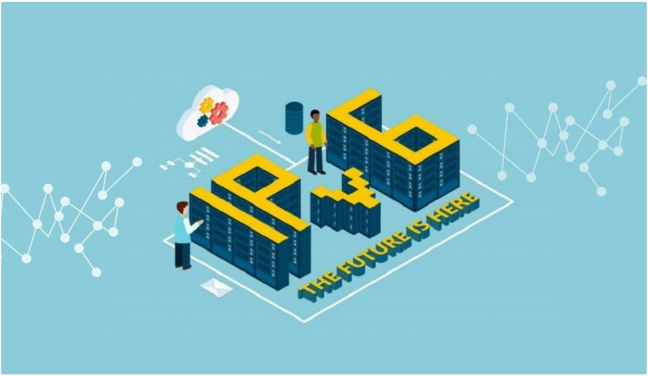
Programming IPv6 Protocols Using Contiki and Cooja

Contiki is a widely used IoT enabled operating system that is free and open source. Cooja is the Contiki network simulator and is also used to program sensor devices. Read on to learn how IPv6 protocols can be programmed with Contiki and Cooja.



Currently, technology has developed to the extent that various devices and gadgets are able to interact with each other, without human interference. This communication technology is today known as the Internet of Things (IoT). IoT based communication is used in a number of applications, including defence equipment, smart cities, smart offices, highway patrolling, smart toll collection, business communications, satellite televisions, and traffic systems or interconnected Web cams for security. IoT is also known and associated with other terms, including ubiquitous computing (UbiComp), pervasive computing or ambient computing, using which a number of devices and objects are virtually connected for remote monitoring and decision making.

The security aspects of the Internet of Things (IoT)

As so many devices and so much equipment are interconnected via a virtual environment, there are serious issues related to security, privacy and the overall performance of IoT networks. These networks are becoming increasingly vulnerable to attacks from assorted sources.

Different types of attacks are used to control and damage the IoT environment at different layers. The attackers can damage and control the IoT network by sending malicious packets and signals, and infrastructure can be virtually destroyed. Such attacks are the most worrisome, as they affect the entire network.

Denial of Service (DoS) attacks: In a DoS attack, network availability is jammed by the attacker node or malicious packets, by capturing the bandwidth or communication channel. Here, authentic and legitimate users are not able to access the network services. This is a notorious attack that works on the network layer of an IoT based scenario, and gets more dangerous when it becomes a Distributed Denial of Service (DDoS) attack, whereby the attacker or malicious node attacks a network from multiple and different locations.

The Sybil attack: This affects the network layer of a vehicular network a lot. In this attack, the identity of the nodes is manipulated. The malicious node attempts to fabricate its identity by pretending to be a registered or original source node. In a Sybil attack, the attacker node creates assorted vehicles or nodes of the same identity by replication, and forces